SMART CITY PARKING SYSTEM

PROJECT PROPOSAL | DAV SUBJECT

Project Description

Smart City Parking System is a dashboard-based software prototype designed to help drivers and city administrators understand parking-space availability across different areas. It presents the total number of parking spaces, available spaces, occupied spaces and parking usage through a clear visual interface.

The dashboard can also display parking occupancy trends, average parking-search time, vehicles entering and exiting, recent parking activity and usage by zone. These visual summaries help users understand parking conditions quickly and support better planning in a smart city.

Objective

The main objective is to reduce the time spent searching for parking, improve the use of existing parking spaces, reduce unnecessary traffic caused by drivers looking for spaces, and make parking management easier for administrators.

Key Features

- Parking overview showing total, available and occupied spaces.
- Zone-wise parking usage to compare demand across different areas.
- Visual charts for parking occupancy trends and average search time.
- Vehicle entry and exit summary with recent activity records.
- Dashboard panels for parking history, reservations, reports and alerts.
- Administrator-friendly interface for monitoring and updating parking information.

Proposed Implementation

The project can be developed as a hardware-free software prototype. HTML, CSS and JavaScript can be used for the frontend. A Python (Flask) or Node.js backend may be added if required, while sample parking-zone data can be stored in a CSV file or a database. The dashboard can be demonstrated using sample or manually entered data.

Expected Outcome

A working prototype that clearly displays parking-space information, summarizes parking usage and demonstrates how a smart city can make more informed parking-management decisions.

Prototype scope: the dashboard shown on the next page is an illustrative interface. Displayed values are sample values and do not represent live sensor data or hardware integration.

PROJECT DASHBOARD

Smart City Parking System — illustrative prototype interface



Dashboard Sections

- Summary cards: total parking spaces, available spaces, occupied spaces and average search time.
- Analytics: parking occupancy trends, search-time chart and vehicles-entered-versus-exited comparison.
- Live parking view and recent activity: a visual overview of spaces and sample vehicle events.
- Zone usage and impact metrics: parking distribution by zone and illustrative fuel/time-saving indicators.